

Owen Asbridge

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Active Secret Security Clearance

Summary

Senior Mechanical Engineering student at South Dakota Mines pursuing aerospace roles in propulsion, structures, testing, and R&D. Hands-on experience taking projects from concept through build and test, with technical depth in CAD, FEA, CFD, dynamic systems, and mechatronics. Currently working on an inspection robot project through an AFGSC National Security Fellowship.

Education

B.S. Mechanical Engineering

South Dakota Mines

Expected May 2027

Rapid City, SD

- **Minors: Aerospace Engineering, Systems Engineering**
- **GPA: 3.856 / 4.0** Dean's List (every semester, Aug 2023 – Present)
- Scholarships: NASA Space Grant Educational Stipend, SDSGC (2026–27); SD-FIRST S-STEM (2025–27)

Experience

Robotics Engineering – National Security Fellow

Air Force Global Strike Command (AFGSC) / Ellsworth AFB

Jun 2026 – Present

Rapid City, SD

- Sole engineer developing a lightweight inspection robot to replace destructive methods for water-intrusion detection in facilities.
- Ran preliminary hand calculations to size structural and drive components, then developed full SolidWorks part and assembly models covering the complete mechanical sub-system.
- Validated part strength with SolidWorks FEA and iterated designs to meet factor-of-safety targets under worst-case loading.
- Reduced component mass by an average of 27.6% via topology optimization and material re-selection, while preserving strength margins and factor of safety.

Mechanical Engineer Intern

AXE Build LLC

May 2025 – Aug 2025

Sheridan, WY

- Engineered structural steel sunshade designs reusing leftover cutoffs from prior jobs, cutting material cost and scrap waste.
- Produced detailed CAD and manufacturing drawings in Autodesk Revit and AutoCAD, supporting the company's PE and increasing drawing throughput.
- Implemented and deployed an offline digital Safety Data Sheet (SDS) app for job sites, ensuring OSHA compliance and improving on-site safety response.
- Researched and evaluated equipment for a new fabrication facility to support workflow efficiency and scalability.

ME221 Grader – Dynamics of Mechanisms

South Dakota Mines

Aug 2025 – May 2026

Rapid City, SD

- Graded homework and provided written constructive feedback for students on kinematic and dynamic analysis of mechanisms.

Carpentry / Construction Laborer

Jameson Tile and Stone

Mar 2019 – Aug 2024

Rapid City, SD

- Measured, cut, and installed tile and stone on residential and commercial jobs while maintaining tight tolerances and finish quality.
- Performed supporting electrical and plumbing work; set up and tore down job-site equipment safely.

Technical Skills

CAD / Design: SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD, Autodesk Revit

Simulation / Analysis: ANSYS Fluent (CFD), SolidWorks FEA, finite-difference solvers, MATLAB Symbolic Toolbox

Programming / Numerical: MATLAB (symbolic, `fmincon`, `ilaplace`, ODE solvers), Python (finite-difference, bisection optimization, NumPy/SciPy computing), Arduino/C++

Prototyping / Manufacturing: 3D printing (FDM, SLA), CNC milling, CNC plasma cutting, hand and power tools

Electronics / Mechatronics: Arduino (UNO, MEGA), sensors (color, light, HD camera), stepper / servo / DC motor control, power distribution, sensor integration

Documentation / Process: GD&T (ASME Y14.5), LaTeX, technical report writing, Microsoft Office, Google Workspace

Technical Projects

More Project Information on GitHub

Autonomous Pick-and-Place Robot – Team Lead

Spring 2026

Mechatronics – 5-person team; served as team lead

- Led 5-person team to design and build a fully autonomous wheeled robot sorting three colored cubes into matching cups, no vision systems, on a \$100 budget.
- Personally designed and fabricated all mechanical subsystems: ball-screw lifting arm, parallel-gripper claw, and chassis using 3D printing with PLA and PETG.
- Integrated TCS3200 and AS7262 color sensors for line following, and ground-indicator matching; three-cycle pick-and-place in ~65 s.
- Led final integration and troubleshooting; diagnosed PWM pin conflicts and executed a mid-project Arduino UNO → MEGA upgrade.

Artemis II Orion Re-entry Capsule – Thermal Protection System Analysis

Spring 2026

Heat Transfer – Individual project

- Performed full stagnation-point aerothermal heating analysis and TPS thickness optimization for Artemis II lunar-return reentry (~11.2 km/s peak).
- Parsed NASA Artemis II ephemeris for reentry trajectory; evaluated freestream conditions using the US Standard Atmosphere 1976 model.
- Applied Fay–Riddell and Sutton–Graves heat-flux correlations with Rankine–Hugoniot post-shock and fully-catalytic wall assumptions; radiation-equilibrium fixed-point iteration for outer wall temperature.
- Built a 1D explicit finite-difference transient heat conduction solver in Python with bisection search over Avcoat thickness; design thickness = 7.40 cm (FS=1.5), peak outer 2,707 K, peak inner 305 K.

Sideline Drink Mixer

Fall 2024

Sophomore Design – 6-person Team

- Collaborated on a team to develop a functional prototype meeting all design requirements.
- Researched and sourced electronic components; designed custom housing in Autodesk Inventor; manufactured via 3D printing and CNC milling.

Arduino RC Car

Fall 2023

Freshman Design – 5-person Team

- Designed and manufactured a functional RC car using Arduino control, 3D-printed components, and CNC plasma-cut structural elements.
- Engineered the steering mechanism in Autodesk Inventor; produced GD&T-annotated manufacturing drawings; iterated on design based on test results.

Relevant Coursework

Aerospace: Rockets and Propulsion; Aerospace Structures; Aircraft Stability and Controls; Space Flight Mechanics; Applied FEA

Mechanical / Systems: Machine Design; Dynamic Systems; Mechatronics; Fluid Mechanics; Heat Transfer; Thermodynamics I–II; Solid Mechanics I–II; Properties of Materials; Numerical Analysis; Differential Equations; Linear Algebra

Leadership & Activities

Campus Ventures – Servant Leadership Team

Aug 2025 – Present

Leadership team member for a Christian ministry organization at South Dakota Mines.

Wildfire Labs Hackathon – Audit Preparedness App

Fall 2025

Weekend hackathon – 2nd place; concept spun into ongoing business by a teammate

- Built an audit-prep tool for credit unions comparing internal policy against regulatory standards to flag compliance gaps; weekend build using AI-assisted coding (Codex).
- **Placed 2nd;** teammate has taken the concept forward as an active business.